

Lecture on Nash's embedding theorem

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The fundamental theorem of Riemannian geometry often refers to the existence and uniqueness of Levi-Civita connection. But it also refers to the result of John Nash [6] that *any Riemannian manifold is isometric to a smooth submanifold in a Euclidean space*. In particular, Riemannian manifold can be defined as a submanifold of Euclidean space with induced metric tensor.

We will prove Nash's theorem for compact manifolds (B4). This case is leading; the general case requires a sequence of additional elementary statements that is outlined in exercises A4 ???.

In the proof we use the so-called Günther trick — an essential simplification of the original proof that was found by Matthias Günther [2]. We also use several minor improvements found by Deane Yang, Terence Tao, and Ralph Howard [3, 8, 9].

A Induced metric

Choose a smooth n -dimensional manifold Ω ; here and further, smooth means C^∞ .

Recall that a field of bilinear symmetric forms, say g , on the tangent bundle $T\Omega$ is called metric tensor. A metric tensor g is called Riemannian if it is positive; that is, if $g(x, x) > 0$ for any tangent vector $x \neq 0$.

We will consider $\mathbb{R}^q$ with standard Euclidean metric. Recall that derivative of $w: \Omega \rightarrow \mathbb{R}^q$ in direction of a tangent vector $x \in T_p \Omega$ is denoted by

$$xw = dw(x).$$

Let $w: \Omega \rightarrow \mathbb{R}^q$ be a smooth map. We say that g is induced metric tensor for w if

$$g(x, y) = \langle xw, yw \rangle$$

for any two tangent vectors $x, y \in T_p \Omega$ at any point $p \in \Omega$.

Suppose $v, w: \Omega \rightarrow \mathbb{R}^q$ are two smooth maps. Let us write $g = Q(v, w)$ if

$$g(x, y) \equiv \frac{1}{2} \cdot [\langle xv, yw \rangle + \langle xw, yv \rangle].$$

The operator Q is symmetric and bilinear; it takes two smooth maps $v, w: \Omega \rightarrow \mathbb{R}^q$ and spits a metric tensor $g = Q(v, w)$ on Ω .

Note that if v and w are C^k -smooth, then $Q(v, w)$ is C^{k-1} -smooth.

A1. Observation. *A metric tensor g is induced by a smooth map $w: \Omega \rightarrow \mathbb{R}^q$ if and only if $g = Q(w, w)$.*

The components of $g = Q(w, w)$ in a local chart can be written as

$$g_{ij} = Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle.$$

A2. Exercise. Let w be a smooth map from a smooth Riemannian manifold (Ω, g) to the Euclidean space $\mathbb{R}^q$. Show that g is induced by w if and only if w is length-preserving; that is, if

$$\text{length } \gamma = \text{length}(w \circ \gamma)$$

for any curve γ in Ω .

If g is induced by w , then the map $w: (\Omega, g) \rightarrow \mathbb{R}^q$ is often called isometric, but we will not use this term since it is very ambiguous.

Let Ω be a smooth manifold. Given two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$, denote by $w = v_1 \oplus v_2$ the map

$$w: \Omega \rightarrow \mathbb{R}^{q_1+q_2} = \mathbb{R}^{q_1} \oplus \mathbb{R}^{q_2}$$

defined by $w: p \mapsto (v_1(p), v_2(p))$.

A3. Exercise. Show that

$$Q(v_1 \oplus v_2, v_1 \oplus v_2) = Q(v_1, v_1) + Q(v_2, v_2)$$

for any two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$.

Recall that support of a map is the closure of the set of its arguments with nonzero values.

A4. Exercise. Let $v_1, v_2: \Omega \rightarrow \mathbb{R}^q$ be smooth maps defined on a smooth manifold Ω . Suppose v_1 and v_2 have disjoint support. Show that

$$Q(v_1 + v_2, v_1 + v_2) = Q(v_1 \oplus v_2, v_1 \oplus v_2).$$

B Formulation

The following theorem is the main result of this lecture.

B1. Theorem. Let (Ω, g) be a compact Riemannian manifold. Then, for sufficiently large integer q , there is a smooth immersion $w: \Omega \rightarrow \mathbb{R}^q$ such that

$$g = Q(w, w).$$

B2. Exercise. Construct a metric tensor g on the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ such that $g(x, x) \geq 0$ for any tangent vector x , but there is no smooth map $w: \mathbb{T}^2 \rightarrow \mathbb{R}^{10}$ with the induced metric g .

B3. Exercise. Use B_4 together with the Whitney embedding theorem to prove the following stronger version of Theorem B_4 .

B4. Theorem. Let (Ω, g) be a compact Riemannian manifold. Then, for sufficiently large integer q , there is a smooth embedding $w: \Omega \rightarrow \mathbb{R}^q$ such that

$$g = Q(w, w).$$

C Free maps

As before, Ω is a smooth n -dimensional manifold.

Recall that a C^1 map $f: \Omega \rightarrow \mathbb{R}^q$ is called regular if its differential df has rank n at each point. In other words, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all n first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A C^2 map $f: \Omega \rightarrow \mathbb{R}^q$ is called free (or two-regular) if an analogous property holds for first *and second* partial derivatives; that is, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all $n + \frac{n \cdot (n+1)}{2}$ partial derivatives $\partial_i f, \partial_{ij} f$ for $i \leq j$ are linearly independent.

C1. Exercise. Show that there is no free map $\mathbb{S}^2 \rightarrow \mathbb{R}^4$.

C2. Exercise. Let $f_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $f_2: \Omega \rightarrow \mathbb{R}^{q_2}$ be smooth maps defined on a smooth manifold Ω . Suppose f_1 is free; show that so is $f_1 \oplus f_2$.

Let f be a smooth function on a manifold Ω . Denote by E_f the smooth map $\Omega \rightarrow \mathbb{R}^2$, defined by

$$E_f(x) = (\cos[f(x)], \sin[f(x)]).$$

In other words, E_f is the composition of two maps

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell} \mathbb{S}^1 \subset \mathbb{R}^2,$$

where $\ell: \mathbb{R} \rightarrow \mathbb{S}^1$ is the natural covering maps $y \mapsto (\cos y, \sin y)$.

C3. Exercise. Consider the (x, y) -plane $\mathbb{R}^2$. Show that the map

$$E_x \oplus E_y \oplus E_{x+y}: \mathbb{R}^2 \rightarrow \mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$$

is free.

Generalize the statement to maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n \cdot (n+1)}$.

C4. Exercise. Let $f: \Omega \rightarrow \mathbb{R}^s$ be a regular smooth map and $F: \mathbb{R}^s \rightarrow \mathbb{R}^q$ is free. Show that the composition $F \circ f: \Omega \rightarrow \mathbb{R}^q$ is free.

Apply the Whitney embedding theorem and C3 to conclude that any smooth manifold admits a free embedding into a Euclidean space.

D Key lemmas: formulations

Let us denote by $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ the n -dimensional torus. The torus comes with a global system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be also written in these coordinates; so we can (and will) think that metric tensor is smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$; here and further

$$N := \frac{n \cdot (n+1)}{2}.$$

Choose $0 < \alpha < 1$, say $\alpha = \frac{1}{2}$. Denote by $|x - y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of Hölder norms of a function $u: \mathbb{T}^n \rightarrow \mathbb{R}$

$$\begin{aligned} \|u\|_{0,\alpha} &:= \sup_{x \in \mathbb{T}^n} \{ |u(x)| \} + \sup_{x, y \in \mathbb{T}^n} \left\{ \frac{|u(x) - u(y)|}{|x - y|_{\mathbb{T}^n}^\alpha} \right\}, \\ \|u\|_{k,\alpha} &:= \max_{|\mathcal{I}| \leq k} \{ \|\partial_{\mathcal{I}} u\|_{0,\alpha} \}, \end{aligned}$$

where $\mathcal{I} = (i_1, \dots, i_n)$ denotes multi-index; so, $\partial_{\mathcal{I}} := \partial_1^{i_1} \dots \partial_n^{i_n}$ and $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions with finite norm $\| \cdot \|_{k,\alpha}$ is called the (k, α) -Hölder space; it is denoted by $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

D1. Exercise. Let $v \in C^\infty(\mathbb{T}^n, \mathbb{R})$. Show that there is a constant c such that

$$\|u \cdot v\|_{k,\alpha} \leq c \cdot \|u\|_{k,\alpha}$$

for any $u \in C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

Further, let us define $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as the space of functions $u: \mathbb{T}^n \rightarrow \mathbb{R}^q$ with all coordinate functions from $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$. Let

$$\|u\|_{k,\alpha} := \max_i \{ \|u_i\|_{k,\alpha}, \dots, \|u_q\|_{k,\alpha} \},$$

where $u_1, \dots, u_q$ are the coordinate functions of u .

From now on, let us fix a smooth free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and let

$$L(w) := 2 \cdot Q(f, w).$$

D2. Nash's lemma. The linear operator

$$L: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^N)$$

has a right inverse

$$M: C^\infty(\mathbb{T}^n, \mathbb{R}^N) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$\|Mh\|_{2,\alpha} \leq a \cdot \|h\|_{2,\alpha}$$

for some constant a .

D3. Günther's lemma. *There is a symmetric bilinear operator*

$$\tilde{Q}: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \times C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$L\tilde{Q} = Q,$$

$$\|\tilde{Q}(w, w)\|_{2,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha}^2,$$

$$\|\tilde{Q}(w, w)\|_{k,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha} \cdot \|w\|_{k,\alpha} + b_k \cdot \|w\|_{k-1,\alpha}^2$$

for any integer $k \geq 3$ and some fixed constants $b_2, b_3, \dots$

E Perturbation

Recall that $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ is a fixed smooth free map and

$$L(w) := 2 \cdot Q(f, w).$$

E1. Perturbation lemma. *There is $r > 0$ such that the equation*

$$\textcircled{1} \quad L(w) + Q(w, w) = h$$

has a smooth solution $w: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for any smooth metric h such that $\|h\|_{2,\alpha} < r$. Moreover, we can assume that

$$\|w\|_{2,\alpha} \leq C \cdot \|h\|_{2,\alpha},$$

for some constant C .

Suppose that a metric tensor g is induced by f . Note that the equation

$$Q(f + w, f + w) = g + h$$

is equivalent to $\textcircled{1}$. Indeed,

$$\begin{aligned} Q(f + w, f + w) &= Q(f, f) + 2 \cdot Q(f, w) + Q(w, w) \\ &= g + L(w) + Q(w, w). \end{aligned}$$

Therefore the perturbation lemma implies the following.

E2. Corollary. *Let g_0 be induced by a smooth free map $f_0: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and h another metric tensor on $\mathbb{T}^n$. If $\|h\|_{2,\alpha}$ is sufficiently small, then $g_1 = g_0 + h$ is induced by a smooth map $f_1: \mathbb{T}^n \rightarrow \mathbb{R}^q$.*

Proof of E1. Recall that M , a , $\tilde{Q}$, and b_2 are provided by D2 and D3. Set $R = \frac{1}{10 \cdot b_2}$ and $r = \frac{R}{10 \cdot a}$. Assume $\|h\|_{2,\alpha} \leq r$. By D3,

$$\Phi: w \mapsto Mh - \tilde{Q}(w, w)$$

is a contraction map from the closed ball $\bar{B}[0, R] \subset C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ to itself.

Consider the sequence of maps $w_0, w_1, \dots: \mathbb{T}^n \rightarrow \mathbb{R}^q$ such that $w_0 = 0$ and

$$w_{n+1} = \Phi(w_n).$$

Note that w_n is smooth for any n . Since Φ is a contraction, w_n converges in $C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as $n \rightarrow \infty$, say $w_n \rightarrow w$. Further,

$$\|w\|_{2,\alpha} \leq R \quad \text{and} \quad w = Mh - \tilde{Q}(w, w).$$

Since $LM = \text{id}$ and $L\tilde{Q} = Q$, applying L to both sides of the last equation, we get that w solves ❶.

It remains to show that w is smooth. Suppose that

❷ *for an integer k , the sequence $\|w_n\|_{k,\alpha}$ is bounded.*

Since

$$\|w\|_{k,\alpha} \leq \lim_{n \rightarrow \infty} \|w_n\|_{k,\alpha}$$

it follows that $w \in C^{k,\alpha}$. Therefore it remains to prove ❷ for each integer $k \geq 2$. The base case $k = 2$ is already done; let us do the induction step.

Suppose $\|w_n\|_{k-1,\alpha}$ is bounded; in particular, there is a constant c_k such that

$$\|Mh\|_{k-1,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq c_k$$

for any n ; here b_k is from D3.

By the second estimate in D3, we get

$$\begin{aligned} \|w_{n+1}\|_{k,\alpha} &\leq \|Mh\|_{k,\alpha} + b_2 \cdot \|w_n\|_{2,\alpha} \cdot \|w_n\|_{k,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq \\ &\leq c_k + b_2 \cdot R \cdot \|w_n\|_{k,\alpha} \leq \\ &\leq c_k + \frac{1}{10} \cdot \|w_n\|_{k,\alpha}. \end{aligned}$$

In particular,

$$\|w_n\|_{k,\alpha} \leq 2 \cdot c_k \quad \implies \quad \|w_{n+1}\|_{k,\alpha} \leq 2 \cdot c_k$$

for any n . Since $w_0 = 0$, induction on n implies ❷ for k . □

F Proof of Nash's lemma

Proof of D2. Recall that we have global periodic coordinates on $\mathbb{T}^n$.

Choose a metric tensor h on Ω . Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the following identities define uniquely a vector field w on $\mathbb{T}^2 \Omega$ that is spanned by $\partial_i f$, $\partial_{ij} f$

$$\langle \partial_i f, w \rangle = 0, \quad \langle \partial_{ij} f, w \rangle = h_{ij}$$

for all i and j .

Since

$$\begin{aligned} L_{ij}(w) &= 2 \cdot Q_{ij}(f, w) = \\ &= \langle \partial_i f, \partial_j w \rangle + \langle \partial_j f, \partial_i w \rangle = \\ &= \partial_j \langle \partial_i f, w \rangle + \partial_i \langle \partial_j f, w \rangle - 2 \cdot \langle \partial_{ij} f, w \rangle, \end{aligned}$$

it defines the needed right inverse $M: h \mapsto w$. □

Let $f: \Omega \rightarrow \mathbb{R}^q$ be a regular smooth map and $p \in \Omega$. Denote by T_p the n -dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p . Denote by $T_p^\perp$ the normal complement of T_p in $\mathbb{R}^q$. We will denote by $\mathbb{T} \Omega$ and $\mathbb{T}_p^\perp \Omega$ the corresponding bundles over Ω .

Further, denote by T_p^2 the space spanned by all first and second-order partial derivatives $\partial_i f$, $\partial_{ij} f$ at p . Let $N_p = T_p^2 \cap T_p^\perp$.

Note that that if f is free, then T_p^2 and N_p are respectively $(n + \frac{n \cdot (n+1)}{2})$ and $\frac{n \cdot (n+1)}{2}$ -dimensional at any point p . In this case we will denote by $\mathbb{T}^2 = \mathbb{T}^2 \mathbb{T}^n$ and $\mathbb{N} = \mathbb{N} \mathbb{T}^n$ the corresponding bundles over $\mathbb{T}^n$.

Suppose $w = Mh$ as in the proof above. Since $\langle \partial_i f, w \rangle = 0$ for every i , we get the following.

F1. Observation. *The inverse M in Nash's lemma (D2) can be choosen so that $Mh(p) \in N_p$ for any $p \in \mathbb{T}^n$.*

F2. Exercise. *Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ be a free map. Given a vector field v in $\mathbb{T} \mathbb{T}^n$ and a metric tensor h on $\mathbb{T}^n$, show that there is a unique vector field w in $\mathbb{N} \mathbb{T}^n$ such that*

$$2 \cdot Q(f, v + w) = h.$$

In other words, given v , there is a right inverse $M_v: h \mapsto v + w$ of L with w in $\mathbb{N} \mathbb{T}^n$.

G Proof of Günther's lemma

Many symmetric bilinear operators $\tilde{Q}$ satisfy $L\tilde{Q} = Q$. In particular, one can take $\tilde{Q} = MQ$, where M is provided by Nash's lemma (D2). However, this choice of $\tilde{Q}$ does not satisfy the required inequalities in Günther's lemma.

The actual construction of $\tilde{Q}$ relies on extra freedom provided by F2. It says that, given a tangent vector field v we may choose M so that $Mh - v \in N\mathbb{T}^n$.

Recall that we use global system of periodic coordinates on $\mathbb{T}^n$ to define the partial derivatives $\partial_1, \dots, \partial_n$. In particular, we can define the Laplacian

$$\Delta u = \sum_s \partial_{ss} u.$$

The next statement easily follows from the so-called Schauder estimates — a result in elliptic partial differential equations. It could be also obtained directly using Fourier analysis on torus; we omit its proof.

G1. Proposition. *Let $k \geq 2$ be an integer and $0 < \alpha < 1$. Then*

$$(\Delta - 1): C^{k,\alpha}(\mathbb{T}^n) \rightarrow C^{k-2,\alpha}(\mathbb{T}^n).$$

is a bi-Lipschitz isomorphism. In particular, the linear operator

$$(\Delta - 1): u \mapsto \Delta u - u$$

has a Lipschitz inverse

$$\textcircled{1} \quad (\Delta - 1)^{-1}: C^{k-2,\alpha}(\mathbb{T}^n) \rightarrow C^{k,\alpha}(\mathbb{T}^n)$$

The operator $(\Delta - 1)^{-1}$ smooths function — it swallows a $C^{k-2,\alpha}$ -function and spits a $C^{k,\alpha}$ -function.

Note that $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1} \partial_i = \partial_i (\Delta - 1)^{-1}.$$

These two properties of $(\Delta - 1)^{-1}$ will be important in the following proof.

Proof of D3. Recall that $Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle$. Let us show that

$$\textcircled{2} \quad (1 - \Delta)Q_{ij}(w, w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij} , A_i and A_j is a linear combination of the following quadratic forms

$$\textcircled{3} \quad w \mapsto \langle \partial_a w, \partial_b w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle.$$

for all possible indexes a, b, c, d .

Indeed,

$$\begin{aligned}
 (1 - \Delta)Q_{ij}(w, w) &= \langle \partial_i w, \partial_j w \rangle - \\
 &\quad - \sum_s [2 \cdot \langle \partial_{is} w, \partial_{js} w \rangle + \langle \partial_{iss} w, \partial_j w \rangle + \langle \partial_i w, \partial_{jss} w \rangle] = \\
 &= \langle \partial_i w, \partial_j w \rangle - 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle + 2 \cdot \langle \Delta w, \partial_{ij} w \rangle - \\
 &\quad - \partial_i \langle \Delta w, \partial_j w \rangle - \partial_j \langle \Delta w, \partial_i w \rangle.
 \end{aligned}$$

Whence **2** follows.

Choose $A = A_i$ or $A = A_{ij}$ for some i and j . Note that

$$\begin{aligned}
 \textcircled{4} \quad & \|A(w)\|_{0,\alpha} \leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
 & \|A(w)\|_{k-2,\alpha} \leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha}^2 + c_k \cdot \|w\|_{k-1,\alpha}^2.
 \end{aligned}$$

for some fixed constants $c_2, c_3, \dots$. Indeed, it is sufficient to check inequalities **4** for each form in **3**, and the latter is straightforward.

Applying **1**, we get

$$\begin{aligned}
 \textcircled{5} \quad & \|(1 - \Delta)^{-1} A(w)\|_{2,\alpha} \leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
 & \|(1 - \Delta)^{-1} A(w)\|_{k,\alpha} \leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha} + c_k \cdot \|w\|_{k-1,\alpha}^2;
 \end{aligned}$$

the constants $c_2, c_3, \dots$ might be different from those above, but each of them depends only on f .

Further, recall that

$$\textcircled{6} \quad L_{ij}(w) = -2 \cdot \langle \partial_{ij} f, w \rangle + \partial_i \langle \partial_j f, w \rangle + \partial_j \langle \partial_i f, w \rangle;$$

see Section F. Let us define $\tilde{Q}$ as a linear combination of $\partial_i f$ and $\partial_{ij} f$ for all $i \leq j$ such that

$$\begin{aligned}
 \textcircled{7} \quad & -2 \cdot \langle \partial_{ij} f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1} A_{ij}(w), \\
 & \langle \partial_i f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1} A_i(w)
 \end{aligned}$$

for all i and j . Since f is free, $\tilde{Q}$ is well defined. Combining **6**, **7**, and **2**, we get

$$L\tilde{Q} = Q.$$

Let $\{e_i, e_{ij}\}_{i \leq j}$ be the dual frame to $\{\partial_i f, \partial_{ij} f\}_{i \leq j}$ in T^2 . Note that

$$\tilde{Q}(w, w) = \sum_i (\Delta - 1)^{-1} A_i(w) \cdot e_i - \frac{1}{2} \cdot \sum_{i \leq j} A_{ij}(w) \cdot e_{ij}.$$

Since the maps $e_i, e_{ij} : T^n \rightarrow \mathbb{R}^q$ are smooth, **5** and D1 imply the last inequality in the lemma for each integer $k \geq 3$. $\square$

H Nash's twist

In this section we prove the following statement which can be regarded as an approximate version of our main theorem (B4).

H1. Proposition. *Let g be a Riemannian metric on $\mathbb{T}^n$. Then there is a metric tensor h on $\mathbb{T}^n$ and one-parameter family of smooth maps $w_t: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for $t > 0$ such that*

$$Q(w_t, w_t) = g + t \cdot h$$

for any t .

It might be tempting to pass to the limit of w_t as $t \rightarrow 0$, but, as you will see, this limit is constant; so it does not solve B4. The main player in the proof of the proposition is the so-called Nash's twist which are about to define.

Let φ and f be smooth functions on a smooth manifold Ω . Given $r > 0$, denote by $\mathbb{S}_r^1$ the circle of radius r in $\mathbb{R}^2$. The Nash's twist $F: \Omega \rightarrow \mathbb{R}^2$ of the triple (r, φ, f) is the following composition

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}_r^1 \xrightarrow{\times \varphi} \mathbb{R}^2,$$

where $\ell_r: \mathbb{R} \rightarrow \mathbb{S}_r^1$ is a length-preserving covering map, say

$$\ell_r(x) = (r \cdot \cos \frac{x}{r}, r \cdot \sin \frac{x}{r}),$$

and $\times \varphi$ denotes the multiplication by φ ; so

$$F(x) := \varphi(x) \cdot (\ell_r \circ f(x)).$$

Recall that $(df)^2$ denotes the metric tensor induced by defined by $f: \Omega \rightarrow \mathbb{R}$; equivalently,

$$(df)^2(x, Y) := df(x) \cdot df(Y) = (xf) \cdot (Yf).$$

H2. Claim. *The metric tensor*

$$\varphi^2 \cdot (df)^2 + r^2 \cdot (d\varphi)^2$$

is induced by Nash's twist F for (r, φ, f) .

Computations. Suppose that x is a tangent vector on Ω , then

$$\begin{aligned} xF &= x(\varphi \cdot \ell_r \circ f) = \\ &= (x\varphi) \cdot (\ell_r \circ f) + \varphi \cdot (\ell'_r \circ f) \cdot (xf). \end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$\langle \mathbf{x}F, \mathbf{y}F \rangle = r^2 \cdot (\mathbf{x}\varphi) \cdot (\mathbf{y}\varphi) + \varphi^2 \cdot (\mathbf{x}f) \cdot (\mathbf{y}f). \quad \square$$

Let $w: \Omega \rightarrow \mathbb{R}^q$ be a smooth map defined on a smooth manifold Ω . Suppose that $w_1, \dots, w_q$ are coordinate functions of w , then the induced metric tensor g can be written as

$$g = Q(v, w) = (dw_1)^2 + \dots + (dw_q)^2.$$

In order to prove B4, we need to find functions $w_1, \dots, w_q: \Omega \rightarrow \mathbb{R}$ that solves the above equation for a given Riemannian metric. The following proposition is a weaker form of B4; it will play a key role in the proof of H1.

H3. Proposition. *Let (Ω, g) be a compact n -dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ on Ω such that*

$$\textcircled{1} \quad g = (\varphi_1)^2 \cdot (df_1)^2 + \dots + (\varphi_q)^2 \cdot (df_q)^2.$$

Proof. The metric tensor g can be written in local cart $(x_1, \dots, x_n)$ as

$$\textcircled{2} \quad g = \sum_{i,j} g_{ij} \cdot dx_i \cdot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$.

Since g is Riemannian, we can choose chart so that ∂_i are orthonormal at a given point p ; in other words, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \neq j$. Since g_{ij} are smooth, given $\varepsilon > 0$, we can choose a neighborhood $U \ni p$ such that

$$\textcircled{3} \quad g_{ii} \leq 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \leq \pm \varepsilon$$

for all $i \neq j$ and any point in U .

Observe that

$$\pm dx_i \cdot dx_j = \frac{1}{2} \cdot (dx_i \pm dx_j)^2 - \frac{1}{2} \cdot (dx_i)^2 - \frac{1}{2} \cdot (dx_j)^2.$$

Plug this formula in $\textcircled{2}$ and take $\textcircled{3}$ into account, assuming that ε is small. We get that g is a linear combination with positive coefficients of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j . In other words, we can take the functions x_i and $x_i \pm x_j$ for all i and j as our functions $f_1, \dots, f_q$ and find $\varphi_1, \dots, \varphi_q$ such that $\textcircled{1}$ holds in a neighborhood of p .

Applying a partition-of-unity argument, we get the global statement. $\square$

In the presented proof, it is sufficient to take $2 \cdot n^2$ functions in each chart; so, if Ω is covered by m charts, then one can take $q = 2 \cdot m \cdot n^2$ functions. However, as it is stated in the following exercise, one may take q independent of m .

H4. Exercise. *Show that q in H3 can be bounded explicitly in terms of n ; say, we can assume that $q = 100 \cdot n^{100}$.*

Try to generalize H3 to noncompact manifolds.

Proof of H1. Suppose that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by H3. Let $r = \sqrt{t}$; denote by F_i the Nash's twist for the triple (r, φ_i, f_i) . By H2,

$$w_t = F_1 \oplus \dots \oplus F_q$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_q)^2$. $\square$

H5. Advanced exercise. *Show that H1 holds for noncompact connected manifolds. Moreover, given any positive smooth function f on Ω , we can assume that h is bounded by $f \cdot g$; that is, $0 \leq h(x, x) \leq f \cdot g(x, x)$ for any tangent vector x on Ω .*

I Reduction to torus

I1. Exercise. *Let Ω be a submanifold in the torus $\mathbb{T}^n$. Show that for any Riemannian metric on Ω is a restriction of a Riemannian metric on $\mathbb{T}^n$.*

J Proof assembling

Proof of B4. By the Whitney embedding theorem, any smooth n -dimensional manifold Ω admits an immersion ι into $\mathbb{R}^{2 \cdot n+1}$. Since Ω is compact, we can assume in addition that ι is short and its image lies in a small ball; if not, then we can pass to the rescaling of $\varepsilon \cdot \iota$ for small $\varepsilon > 0$. Composing ι with the covering map $\mathbb{R}^{2 \cdot n+1} \rightarrow \mathbb{T}^{2 \cdot n+1}$, we get an embedding $\Omega \hookrightarrow \mathbb{T}^{2 \cdot n+1}$.

In particular, we can assume that Ω is a compact submanifold of a torus. By I1, any Riemannian metric g on Ω is the restriction of a Riemannian metric on the torus. Therefore,

❶ *it is sufficient to prove the theorem for $\Omega = \mathbb{T}^n$.*

Choose a free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$; it exists by C3. Suppose g_1 is the metric tensor induced by f . We can assume that $g_1 < g$; that is, $g_1(X, X) < g(X, X)$ for any tangent vector $X \neq 0$. In particular, $g_2 = g - g_1$ is a Riemannian metric.

Applying K2, we get a metric tensor h on $\mathbb{T}^n$ such that for any $t > 0$ there is a smooth map $f_2: \mathbb{T}^n \rightarrow \mathbb{R}^{q_2}$ that induces $g_2 + t \cdot h$.

By the perturbation lemma, for any small $t > 0$ there is a smooth map $w: \mathbb{T}^n \rightarrow \mathbb{R}^{q_1}$ such that

$$Q(f_1 + w, f_1 + w) = g_1 - t \cdot h.$$

Finally by A3, the metric tensor

$$g = g_1 - t \cdot h + g_2 + t \cdot h$$

is induced by the map $(f_1 + w) \oplus f_2$. □

J1. Exercise. *Let Ω be a connected noncompact manifold. Show that it admits a smooth proper bounded embedding in $\mathbb{R}^q \setminus \{0\}$. Moreover, given a Riemannian metric g on Ω , we can assume that the induced metric of this embedding is strictly smaller than g .*

K Remarks

Let us state another closely related result that shows a huge difference between C^1 and C^2 isometric embeddings. For example, it implies that the unit sphere admits C^1 length-preserving embedding into an arbitrarily small ball in Euclidean 3-space. There is no such C^2 -embedding since Gauss curvature of the unit sphere is 1, but at an extremal point it must be at least $\frac{1}{r^2}$, where r is the radius of the ball.

K1. Nash–Kuiper theorem. *Let (Ω, g) be a n -dimensional Riemannian manifold and $f: (\Omega, g) \rightarrow \mathbb{R}^q$ be a short smooth regular map. Suppose that $q \geq n + 1$. Then for any $\varepsilon > 0$ there is an C^1 -smooth length-preserving maps $f_\varepsilon: (\Omega, g) \rightarrow \mathbb{R}^q$ that is ε close to f ; that is, $|f_\varepsilon(x) - f(x)| < \varepsilon$ for any $x \in \Omega$.*

Moreover if f is an embedding then we can assume that so is f_ε .

It was originally proved by John Nash [5] with the condition $q \geq n + 2$ and improved to $q \geq n + 1$ by Nicolaas Kuiper [4]. The original proof uses Nash's twist in a different way. Both papers are reader-friendly. One may also start with lectures by Allan Yashinski and the author [7] where related results were obtained using an alternative approach.

The discussed result formed a part of foundations of the so-called homotopy principle, or h-principle; an excellent introduction is given in the book by Yakov Eliashberg and Nikolai Mishachev [1].

Many related questions are open. For example, it is unknown if *a neighborhood of any point in 2-dimensional Riemannian manifold admits a smooth length-preserving embedding into $\mathbb{R}^3$.*

This theorem is a leading partial case of Nash theorem on regular embedding. The compactness condition can be removed; we will give a hint how to do it. Moreover the value q can be estimated in terms $n = \dim \Omega$. We will give a hint how to get a ruf bound, say $q = 100 \cdot n^3$. The best known estimate so far is

$$q = \max\left\{\frac{n \cdot (n+5)}{2}, \frac{n \cdot (n+3)}{2} + 5\right\}.$$

In the following exercises we assume that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by H3 for a given Riemannian manifold (Ω, g) . Further, F_i is the Nash's twist for the triple (r, φ_i, f_i) , and

$$\begin{aligned} h &= (d\varphi_1)^2 + \dots + (d\varphi_q)^2, \\ F_r &= F_1 \oplus \dots \oplus F_q: \Omega \rightarrow \mathbb{R}^{2 \cdot q}, \\ \Phi &= \varphi_1 \oplus \dots \oplus \varphi_q: \Omega \rightarrow \mathbb{R}^q. \end{aligned}$$

K2. Exercise. *Let (Ω, g) be a compact n -dimensional Riemannian manifold and $r > 0$.*

- (a) *Show that $g + r^2 \cdot h$ is induced by F_r .*
- (b) *Show that $F_r(p) \rightarrow 0$ as $r \rightarrow 0$ for any point $p \in \Omega$.*

Let us denote by $\mathbb{R}^{r,s}$ the pseudoeuclidean space with signature (r, s) ; that is, the space $\mathbb{R}^{r+s}$ with scalar product defined by

$$\langle x, y \rangle = x_1 \cdot y_1 + \dots + x_s \cdot y_s - x_{s+1} \cdot y_{s+1} - \dots - x_{s+r} \cdot y_{s+r},$$

where x_i and y_i denote the coordinates of vectors x and y in $\mathbb{R}^{r+s}$.

Show that g is induced by

$$F_r \oplus \Phi: \Omega \rightarrow \mathbb{R}^{2 \cdot q, q} = \mathbb{R}^{2 \cdot q, 0} \oplus \mathbb{R}^{0, q}.$$

The presented construction can be used to find an isometric immersion of Riemannian manifold (Ω, g) into a pseudoeuclidean space $\mathbb{R}^{r+s}$ with sufficiently large r and s .

K3. Exercise. *Let $f: (\Omega, g) \rightarrow \mathbb{S}^{q-1}$ be a smooth length-preserving embedding. Construct a smooth length-preserving embedding of any conformally equivalent manifold $(\Omega, \varphi^2 \cdot g)$ into $\mathbb{R}^{q,1}$.*

That is, given a smooth positive function φ on Ω , construct a smooth map $\Omega \rightarrow \mathbb{R}^{q,1}$ with induced metric tensor $\varphi^2 \cdot g$.

Semisolutions

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